CIMA Management Case Study

Mock Exam 3

Cartn

Unseen [3 hours]

Task 1 - Unseen case material [45 minutes]

You received the following email:

From: Elizabeth Maenda, Senior Financial Manager
To: Financial Manager
Subject: Fruitello Recall and Corema Industries Supply Terms

Hi,

I trust you have seen the attached news regarding the Fruitello product recall. The Board is concerned about the implications for our own consultancy and supply arrangements.

I have attached a Briefing Note from the Operations Director which provides more detail on the situation and some issues we've identified with one of our own key suppliers. I need your advice on the following:

I would be grateful if you could assist with the following:

- Identify and evaluate the risks to Cartn arising from product/service reputation and contractual inadequacy in relation to its design and advisory services.

[sub-task (a) = 60%]

- Recommend with reasons the approach the company could take to renegotiate terms with one of our major suppliers.

[sub-task (b) = 40%]

Kind regards,
Elizabeth

Reference material

News article

HARRLAND BUSINESS NEWS

Fruitello issues large-scale product recall following packaging concerns

Fruitello, a major fruit juice brand, has issued a large-scale product recall affecting hundreds of thousands of 200ml and 500ml fruit juice cartons after consumers reported an unusual taste and odour in their products.

Investigations suggest the incident may be linked to a design flaw in the packaging, with concerns that the polyethylene layer used in the carton's inner lining may not have been correctly specified in terms of its thickness and grade. This allegedly resulted in inadequate separation between the aluminium layer and the acidic fruit juice contents, potentially tainting the product and raising consumer safety concerns. The specifications were developed by the packaging manufacturer during the design process for Fruitello, highlighting the role of external expertise in packaging development.

Fruitello initiated the recall after identifying the issue, and the packaging manufacturer has confirmed it is conducting an internal review of its design processes. Industry observers suggest that legal proceedings or claims could follow.

The incident has also raised broader considerations around how responsibilities are defined across the packaging design process, particularly where external parties such as design or advisory specialists are involved. In such situations, contractual clarity can play an important role in determining how issues are addressed and where responsibility ultimately lies.

Internal briefing note

To: The Board
From: Alison Cruikshank, Operations Director
Subject: Internal Review: Laminar Solutions and Corema Industries

Laminar Solutions (Fruitello Incident)

Industry intelligence confirms that the unnamed manufacturer in the Fruitello recall is Laminar Solutions. Like Cartn, they are a major rival providing both manufacturing and consultancy services. This incident highlights the significant liability risks associated with providing technical design specifications to consultancy clients.

Supply Chain Review: Corema Industries

Following the news, we have conducted an urgent review of our contract with Corema Industries, a major polyethylene supplier. Two areas of concern have been identified:

- Liability Clauses: These are currently too vague, creating ambiguity over financial responsibility in the event of material failure.
- Quality Specifications: The contract does not clearly define the required grades or technical standards for materials supplied.

Opportunity for Renegotiation

Corema Industries has recently requested a meeting to revise their current pricing and supply volumes. The Board intends to use this as a trigger to renegotiate the entire contract to better protect Cartn's interests.

Task 2 - Unseen case material [45 minutes]

You received the following email:

From: Elizabeth Maenda, Senior Financial Manager
To: Financial Manager
Subject: HoloScan

Hi,

Please see the email below from Wofai Eko regarding HoloScan, a holographic visualisation technology that Wofai is proposing for application in our packaging design and advisory services.

I would be grateful if you could assist with the following:

- Evaluate the arguments both for and against regarding HoloScan as a disruptive technology.

[sub-task (a) = 40%]

- Explain the key considerations when developing non-financial performance indicators to assess the value generated by the proposed HoloScan investment, focusing on areas such as customer and market impact, operational performance, and long-term development.

[sub-task (b) = 60%]

Kind regards,
Elizabeth

Reference material

Email

From: Wofai Eko, Director of Product Development
To: All Board Members
Cc: Senior Managers
Subject: HoloScan – Holographic Visualisation Technology for Packaging Design

Dear All,

Following the Board's request to identify solutions to reduce the risk of errors in our product design and advisory services, I am pleased to present HoloScan as a potential response to this challenge.

HoloScan is a holographic visualisation and 3D modelling technology widely adopted in the aerospace industry, where it is used to create precise holographic representations of components before manufacture. Engineers use it to identify design flaws, material incompatibilities, and structural weaknesses early, reducing costly errors in safety-critical applications.

In the context of our advisory services, HoloScan would allow our consultants to help customers visualise their requested packaging design in 3D, enabling collaborative review and refinement of specifications before any physical work begins. This would reduce the risk of errors arising from miscommunication or unclear specifications, improve customer satisfaction, and minimise costly design iterations.

The technology could also be used internally to simulate how different material layers interact under various conditions. For example, it could test how different polyethylene grades and thicknesses interact with other carton layers, ensuring specifications are technically sound before production begins.

Whilst new to the food packaging industry, HoloScan's underlying technology is well established. I would welcome the Board's support in exploring this further.

Kind regards,
Wofai

Task 3 - Unseen case material [45 minutes]

You received the following email:

From: Elizabeth Maenda, Senior Financial Manager
To: Financial Manager
Subject: HoloScan Investment – Financing Options

Hi,

Following the Board's interest in progressing the HoloScan proposal, Finance Director Mario Bacquero has asked that we assess the most appropriate way to fund the investment, which is estimated to cost approximately H$5 million.

Two financing options have been identified for consideration. Please find attached a summary of the key terms of each option.

I would be grateful if you could assist with the following:

- Identify the characteristics of debt and equity that affect their suitability for the funding of the HoloScan investment and recommend the more suitable source of funds, stating reasons.

[sub-task (a) = 60%]

- Evaluate how the funding of this investment will affect key accounting ratios.

[sub-task (b) = 40%]

Kind regards,
Elizabeth

Reference material

Financing Options and Ratios Summary - HoloScan Investment

Option 1 – Bank Loan

Cartn has received indicative terms from two lenders offering the following:

	Lender A	Lender B
Amount	H$5 million	H$5 million
Interest rate	Fixed at 6% per annum	Variable, currently at 4% per annum
Term	5 years	5 years
Repayment	Monthly	Monthly

Both facilities would be secured against existing assets. Cartn's long-term loans stand at H$154.3 million.

Option 2 – Rights Issue

Cartn could raise H$5 million through a rights issue, offering existing shareholders the right to subscribe for new shares at a discount to the current market price. The rights issue would be offered on a pro-rata basis to existing shareholders, preserving their proportionate ownership in the company.

Current Ratios for Reference

	2026	2025
Current ratio	2.04	1.91
Gearing	28.5%	31.4%
Interest cover	11.4	7.1
ROCE	20.7%	14.12%

Task 4 - Unseen case material [45 minutes]

You received the following email:

From: Elizabeth Maenda, Senior Financial Manager
To: Financial Manager
Subject: HoloScan – Pricing and Customer Communication

Hi,

Please see the email below from Haitham Amer regarding the consultants' concerns around HoloScan and its proposed pricing. Finance Director Mario Baquero and I would like your assistance with the following:

- Evaluate the benefits and drawbacks of bundled pricing and recommend the approach that should be taken to price consulting services after the implementation of HoloScan.

[sub-task (a) = 40%]

- Identify customer needs and recommend how consultants should communicate to customers the decision to use HoloScan in design consultations, including any associated additional costs.

[sub-task (b) = 60%]

Kind regards,
Elizabeth

Reference material

Email

From: Haitham Amer, Marketing Director
To: Mario Bacquero, Finance Director
Cc: Elizabeth Maenda, Senior Financial Manager
Subject: HoloScan – Customer Acceptance and Pricing Concerns

Dear Mario, Elizabeth,

I am writing to raise some concerns that have emerged from conversations with our consultants regarding the proposed implementation of HoloScan in our design consultations.

Whilst our consultants recognise the technical benefits of HoloScan, they have expressed concern that some customers may be resistant to its use during the consultation and design process. Customers who are accustomed to traditional design methods may be unfamiliar with holographic visualisation technology and may question its relevance or reliability in the context of food packaging design.

A related concern is around pricing. Our consultancy fees are currently based on estimated time and engineer seniority, ranging from approximately H$150 to H$400 per hour. I understand the Finance Department is considering a bundled pricing approach to recover the HoloScan investment. This would absorb the technology cost into a revised range of approximately H$180 to H$450 per hour, rather than itemising it separately.

Our consultants are unsure how to communicate this change. They worry that failing to explicitly inform customers of the basis for these higher fees could create resistance or jeopardise existing relationships. We need to develop a clear and consistent approach to both pricing and customer communication before the rollout.

Kind regards,
Haitham

Present value tables

Present value of U$1 i.e. $(1 + r)^{-n}$ where r = interest rate, n = number of periods until payment or receipt.

Periods (n)	Interest rates (r) 1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.962	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.905	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.788	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149

Periods (n)	Interest rates (r) 11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.901	.893	.885	.877	.870	.862	.855	.847	.840	.833
2	.812	.797	.783	.769	.756	.743	.731	.718	.706	.694
3	.731	.712	.693	.675	.658	.641	.624	.609	.593	.579
4	.659	.636	.613	.592	.572	.552	.534	.516	.499	.482
5	.593	.567	.543	.519	.497	.476	.456	.437	.419	.402
6	.535	.507	.480	.456	.432	.410	.390	.370	.352	.335
7	.482	.452	.425	.400	.376	.354	.333	.314	.296	.279
8	.434	.404	.376	.351	.327	.305	.285	.266	.249	.233
9	.391	.361	.333	.308	.284	.263	.243	.225	.209	.194
10	.352	.322	.295	.270	.247	.227	.208	.191	.176	.162
11	.317	.287	.261	.237	.215	.195	.178	.162	.148	.135
12	.286	.257	.231	.208	.187	.168	.152	.137	.124	.112
13	.258	.229	.204	.182	.163	.145	.130	.116	.104	.093
14	.232	.205	.181	.160	.141	.125	.111	.099	.088	.078
15	.209	.183	.160	.140	.123	.108	.095	.084	.074	.065
16	.188	.163	.141	.123	.107	.093	.081	.071	.062	.054
17	.170	.146	.125	.108	.093	.080	.069	.060	.052	.045
18	.153	.130	.111	.095	.081	.069	.059	.051	.044	.038
19	.138	.116	.098	.083	.070	.060	.051	.043	.037	.031
20	.124	.104	.087	.073	.061	.051	.043	.037	.031	.026

Cumulative present value tables

This table shows the Present Value of U$1 per annum, Receivable or Payable at the end of each year for n years

$$\frac{1-(1+r)^{-n}}{r}$$

Periods (n)	Interest rates (r) 1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	8.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	17.226	15.679	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	18.046	16.351	14.878	13.590	12.462	11.470	10.594	9.818	9.129	8.514

Periods (n)	Interest rates (r) 11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0685	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870